Blockchain Lab
EXP 4

Aim - Hands on Solidity Programming Assignments for creating Smart Contracts.

Theory -

1. **Primitive Data Types, Variables, Functions – pure, view**

In Solidity, primitive data types form the foundation of smart contract development. Commonly used types include:

- **uint / int**: unsigned and signed integers of different sizes (e.g., uint256, int128).
- **bool**: represents logical values (true or false).
- **address**: holds a 20-byte Ethereum account address, often used for storing user accounts or contract addresses.
- **bytes / string**: store binary data or textual data.

Variables in Solidity can be **state variables** (stored on the blockchain permanently), **local variables** (temporary, created during function execution), or **global variables** (special predefined variables such as msg.sender, msg.value, and block.timestamp).

Functions allow execution of contract logic. Special types of functions include:

- **pure**: cannot read or modify blockchain state; they work only with inputs and internal computations.
- **view**: can read state variables but cannot alter them. This classification helps optimize gas usage and enforces function integrity.

2. **Inputs and Outputs to Functions**

Functions in Solidity can accept input arguments and return one or more output values. Inputs enable users or other contracts to pass data into the contract, while outputs make it possible to return results after computation. For example, a function can accept an amount in Ether and return whether the transfer was successful. Solidity also allows named return variables, which improve readability and debugging.

3. Visibility, Modifiers and Constructors

- **Function Visibility** defines who can access a function:
    - public: available both inside and outside the contract.
    - private: only accessible within the same contract.
    - internal: accessible within the contract and its child contracts.
    - external: can be called only by external accounts or other contracts.

- **Modifiers** are reusable code blocks that change the behavior of functions. They are often used for access control, such as restricting sensitive functions to the contract owner (onlyOwner).
- **Constructors** are special functions executed only once during contract deployment. They initialize important values, such as setting the deploying account as the owner of the contract.

3. Control Flow: if-else, loops

Control flow in Solidity is similar to traditional programming languages:

- **if-else** allows conditional decision-making in contract logic, e.g., checking if a balance is sufficient before transferring funds.
- **Loops** (for, while, do-while) enable repeated execution of code. For example, iterating through an array of users. However, loops must be used carefully, as excessive iterations increase gas consumption, potentially making the contract expensive to execute.

5. Data Structures: Arrays, Mappings, Structs, Enums

- **Arrays**: Can be fixed or dynamic and are used to store ordered lists of elements. Example: an array of addresses for registered users.
- **Mappings**: Key-value pairs that allow quick lookups. Example: mapping(address => uint) for storing balances. Unlike arrays, mappings do not support iteration.
- **Structs**: Allow grouping of related properties into a single data type, such as creating a struct Player {string name; uint score;}.
- **Enums**: Used to define a set of predefined constants, making code more readable. Example: enum Status { Pending, Active, Closed }.

6. Data Locations

Solidity uses three primary data locations for storing variables:

- **storage**: Data stored permanently on the blockchain. Examples: state variables.
- **memory**: Temporary data storage that exists only while a function is executing. Used for local variables and function inputs.
- **calldata**: A non-modifiable and non-persistent location used for external function parameters. It is gas-efficient compared to memory. Understanding data locations is essential, as they directly impact gas costs and performance.

7. Transactions: Ether and Wei, Gas and Gas Price, Sending Transactions

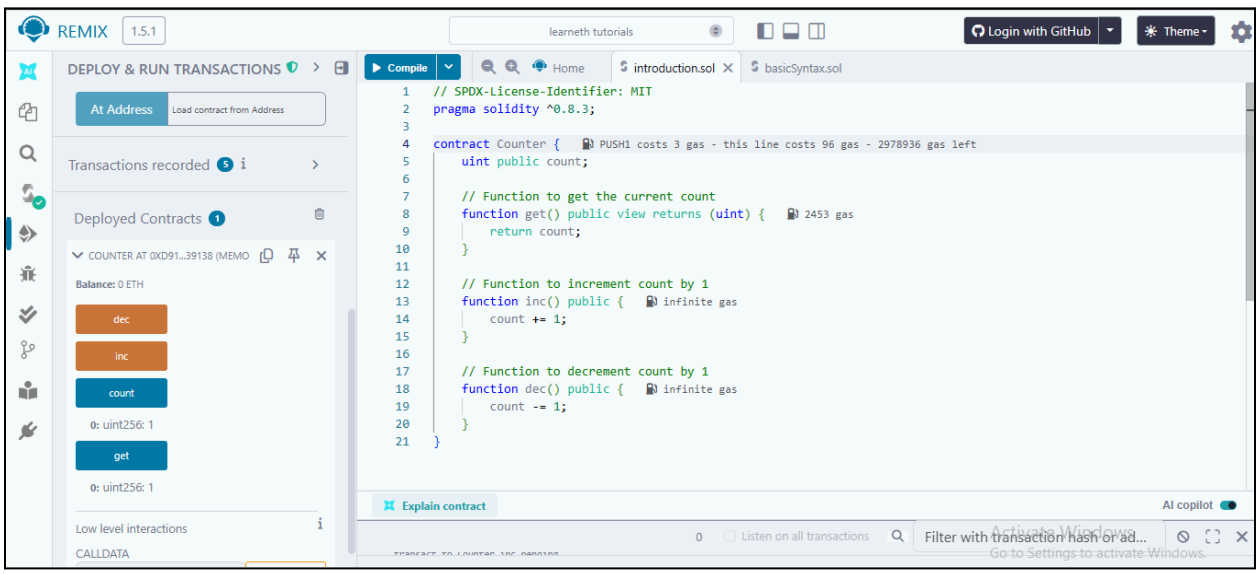
- **Ether and Wei**: Ether is the main currency in Ethereum. All values are measured in Wei, the smallest unit (1 Ether = 10^{18} Wei). This ensures high precision in financial transactions.

- **Gas and Gas Price**: Every transaction consumes gas, which represents computational effort. The gas price determines how much Ether is paid per unit of gas. A higher gas price incentivizes miners to prioritize the transaction.
- **Sending Transactions**: Transactions are used for transferring Ether or interacting with contracts. Functions like transfer() and send() are commonly used, while call() provides more flexibility. Each transaction requires gas, making efficiency in contract design very important.

Code & Output -

Tutorial 1

1. Get counter value



```
from                    0x5B38Da6a701c568545dCfcB03FcB875f56beddC4

to                      Counter.get() 0xd9145CCE52D386f254917e481eB44e9943F39138

execution cost          2453 gas (Cost only applies when called by a contract)

input                   0x6d4...ce63c

output                  0x0000000000000000000000000000000000000000000000000000000000000001

decoded input           {}

decoded output          {
                            "0": "uint256: 1"
                        }
```

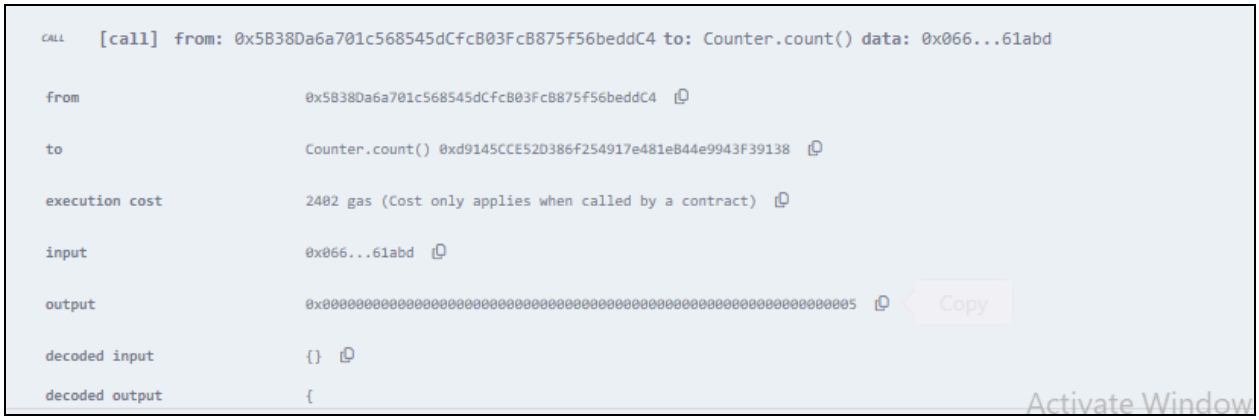
2. Increment counter value

```
[vm] from: 0x5B3...eddC4 to: Counter.inc() 0xd91...39138 value: 0 wei data: 0x371...303c0 logs: 0 hash: 0xfce...8cdc3   Debug

status                  1 Transaction mined and execution succeed
transaction hash        0xfcec1001233a0cc437ac7e87c2db11f10da2fe0976a8ef4e98ee5c6c5228cdc3
block hash              0xc8733a651aaad2a98d67f80a95784141e7e550cd40388c50feefcd1145229228
block number            6
from                    0x5B38Da6a701c568545dCfcB03FcB875f56beddC4
to                      Counter.inc() 0xd9145CCE52D386f254917e481eB44e9943F39138
transaction cost        26417 gas
```

3. Decrement counter value

```
[vm] from: 0x5B3...eddC4 to: Counter.dec() 0xd91...39138 value: 0 wei data: 0xb3b...cfa82 logs: 0 hash: 0x21d...52d6d   Debug

status                  1 Transaction mined and execution succeed
transaction hash        0x21daa184e0a457ef2f35508a0094a8fc0c2eac05b53ab95e6d96e1c2c4452d6d
block hash              0xca425c3b6aa253d68e49726515232861814af84154eb74afe6c4683415457db8
block number            7
from                    0x5B38Da6a701c568545dCfcB03FcB875f56beddC4
to                      Counter.dec() 0xd9145CCE52D386f254917e481eB44e9943F39138
transaction cost        26461 gas
```

4. Get count value



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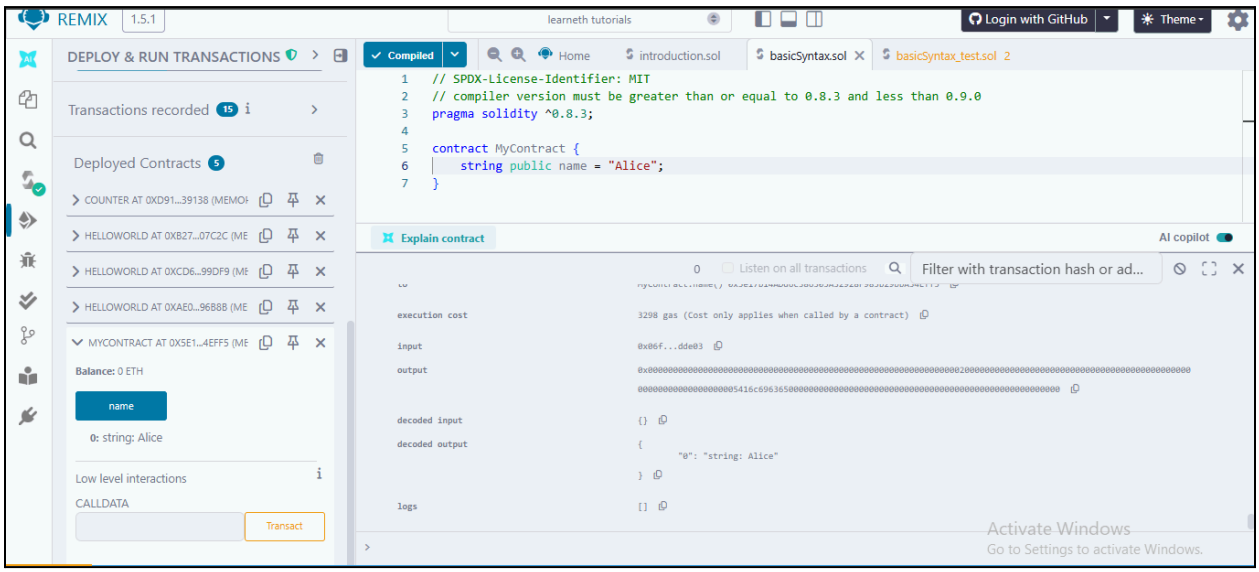

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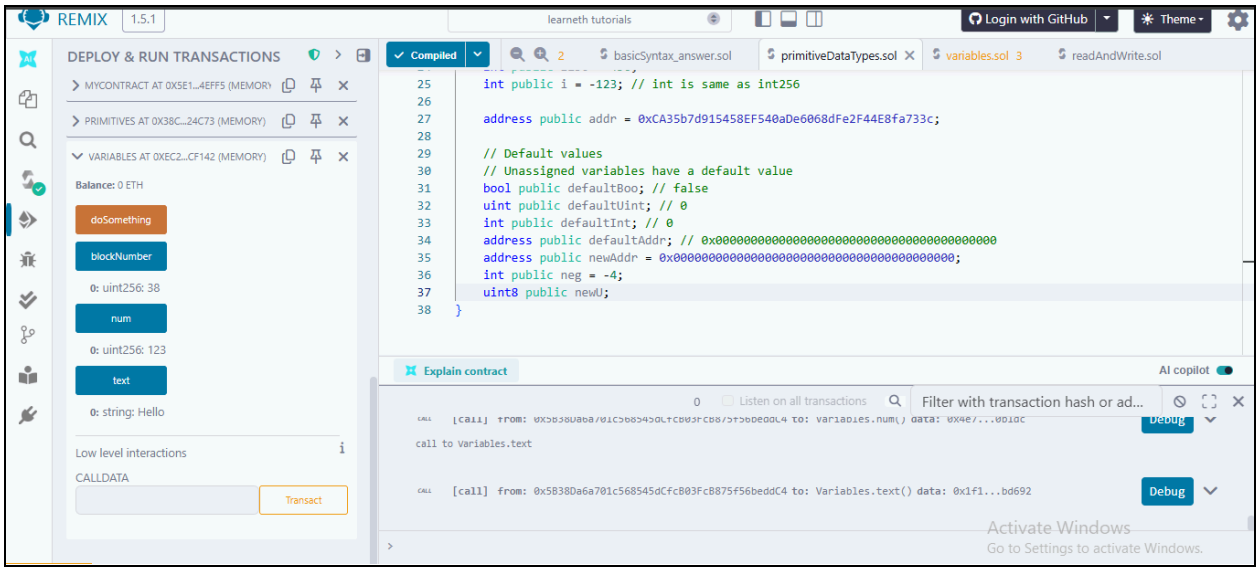

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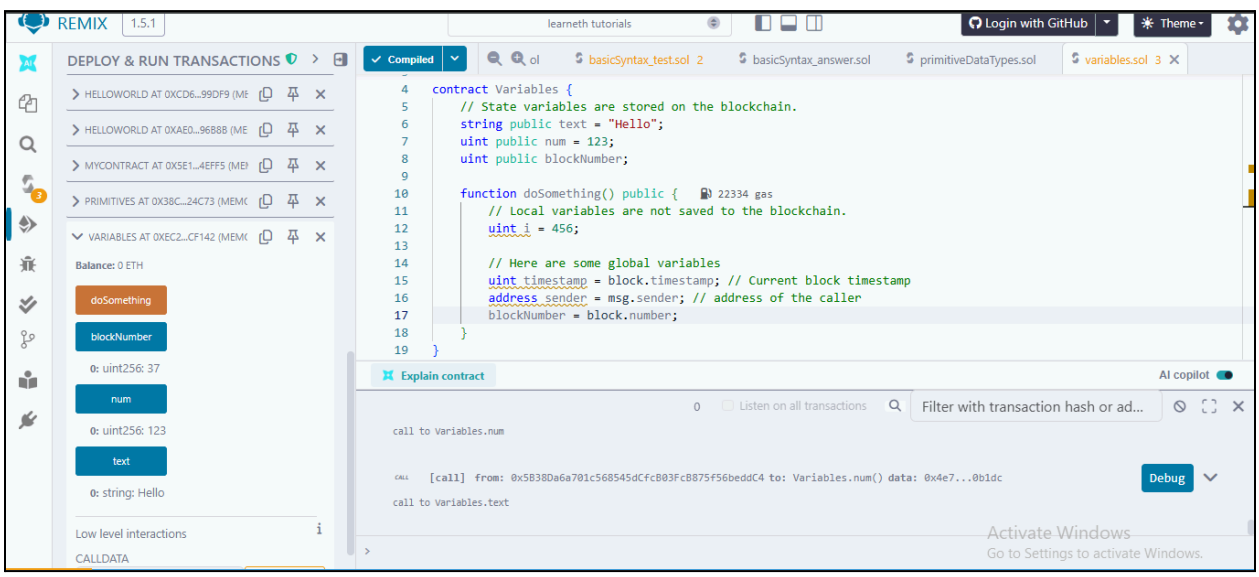

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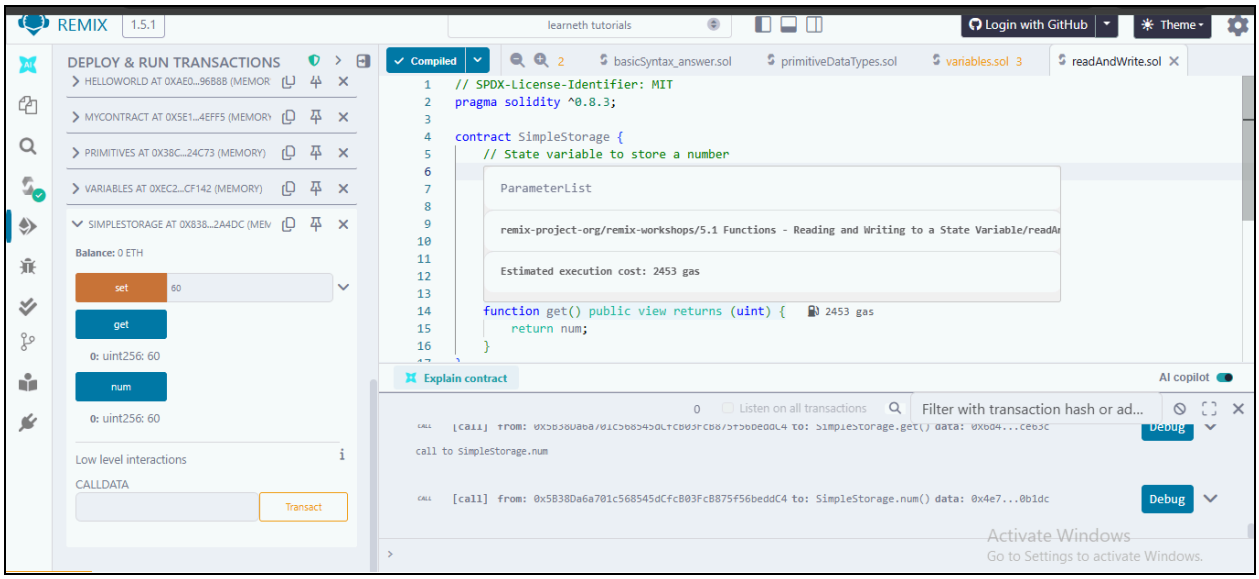

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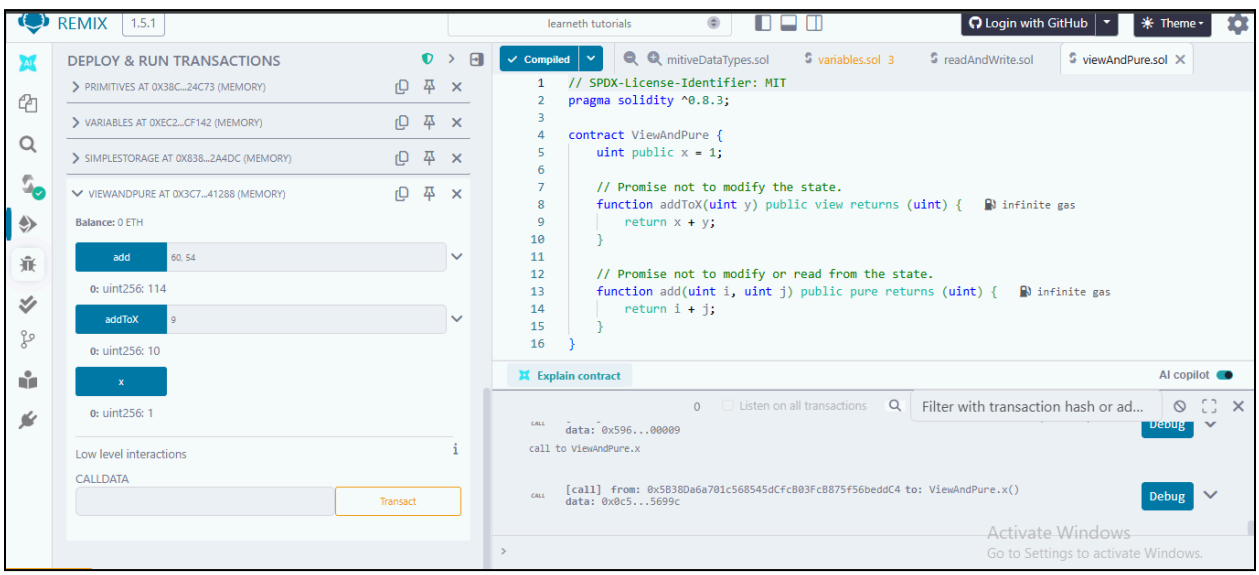

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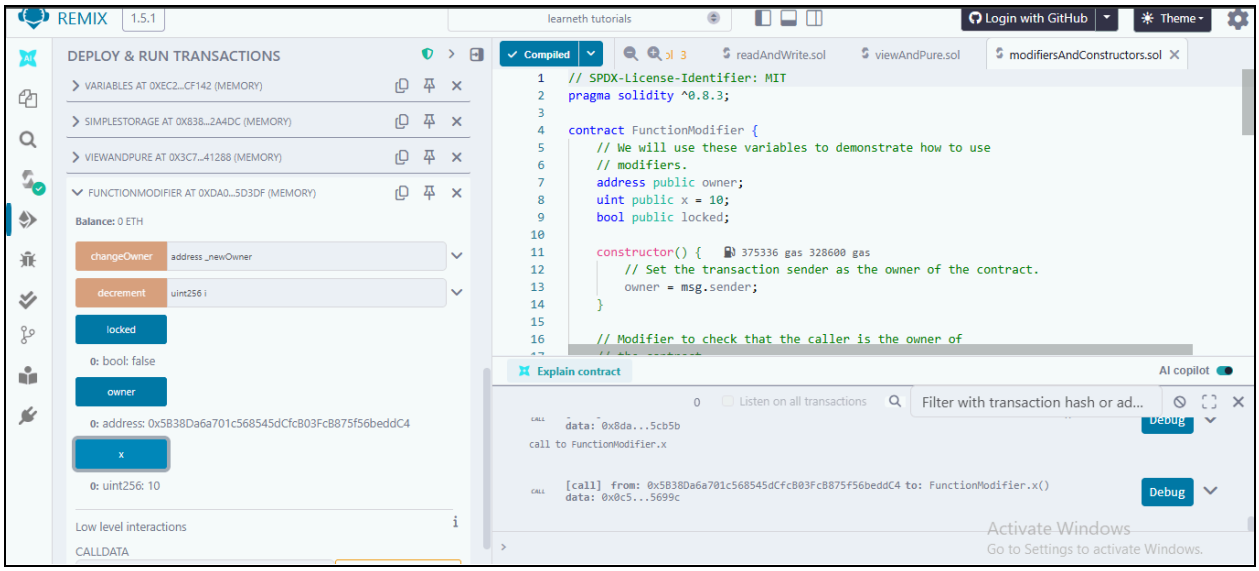

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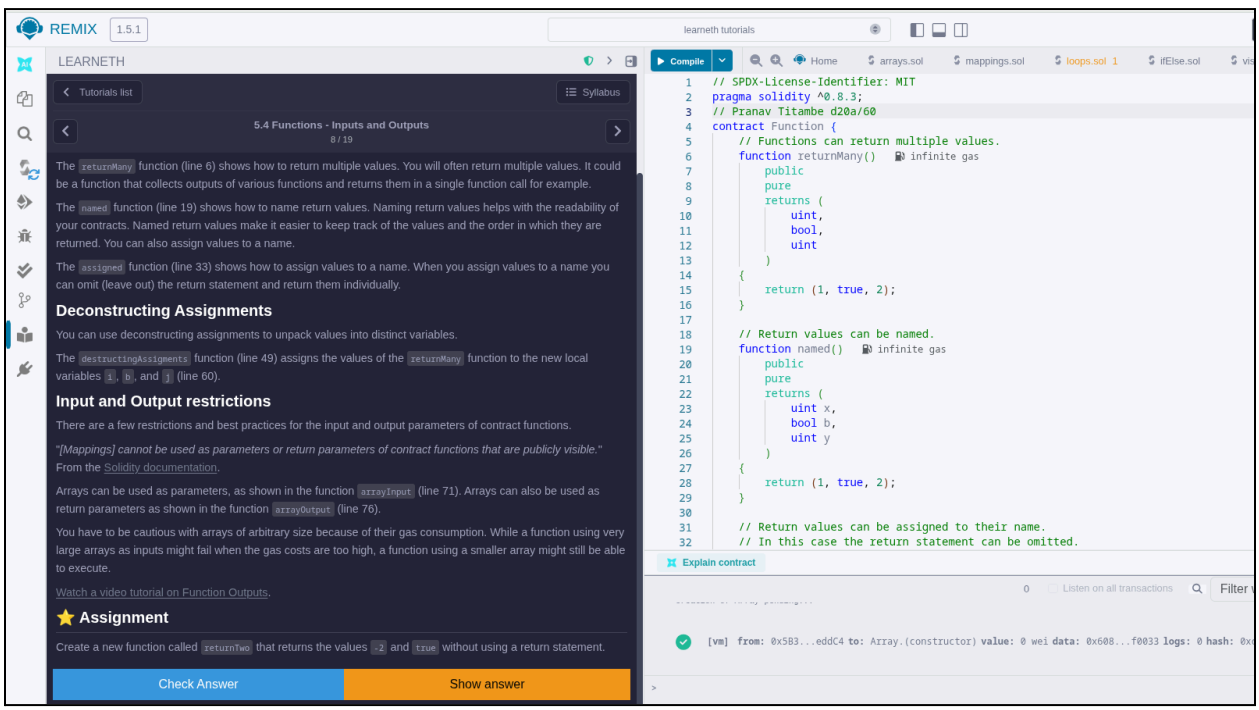

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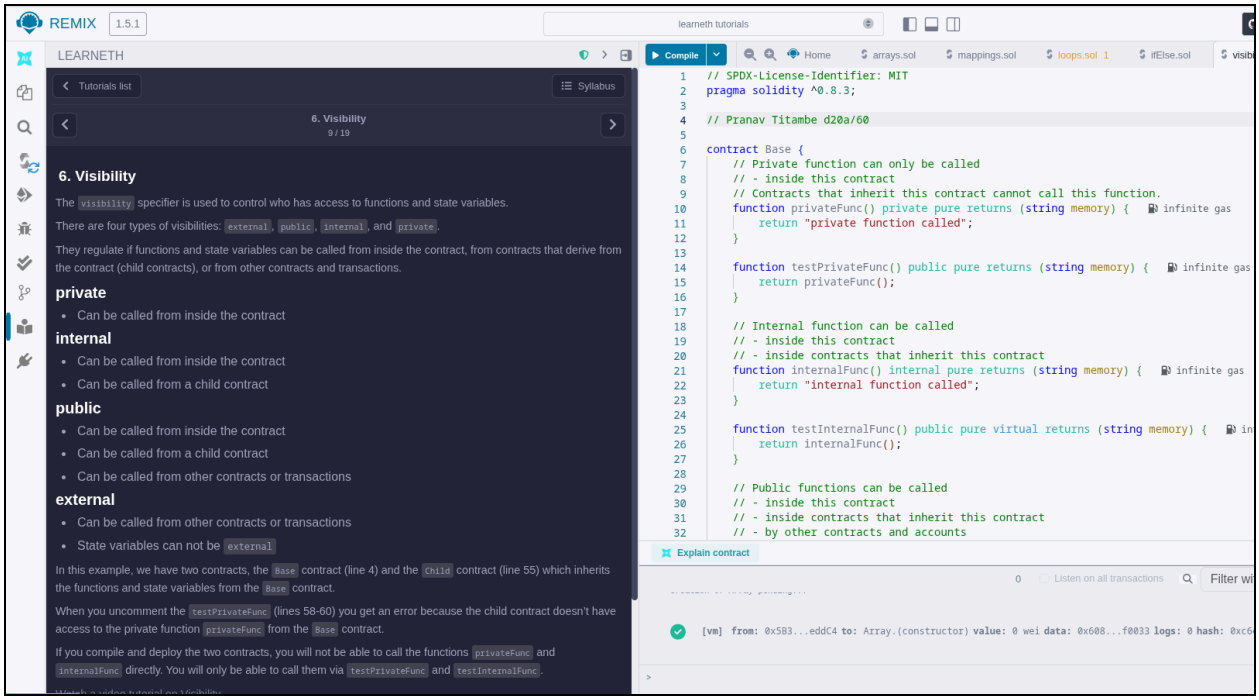

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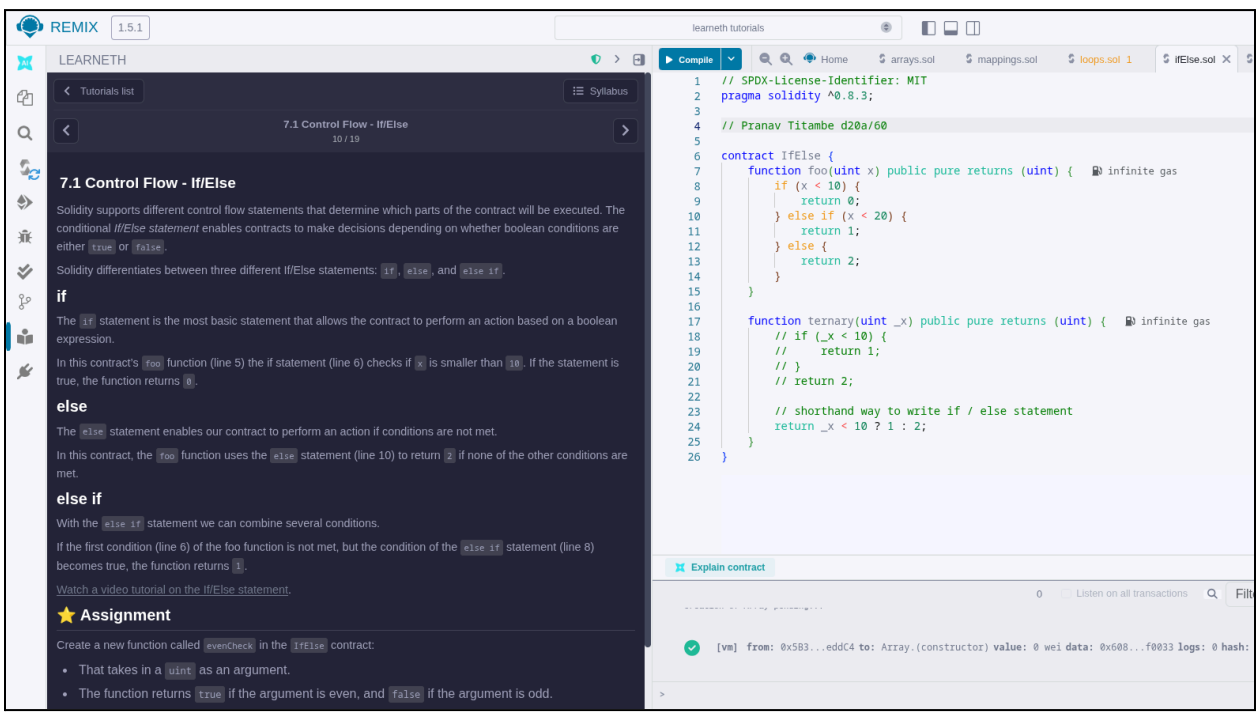

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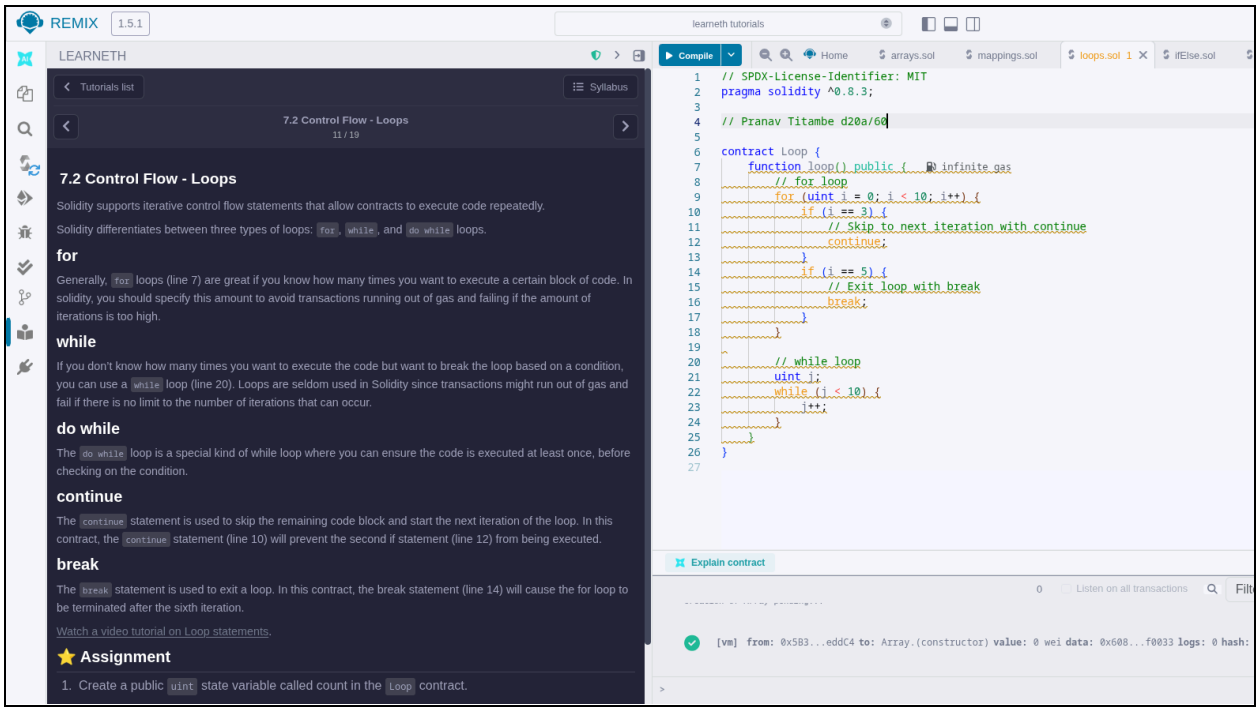

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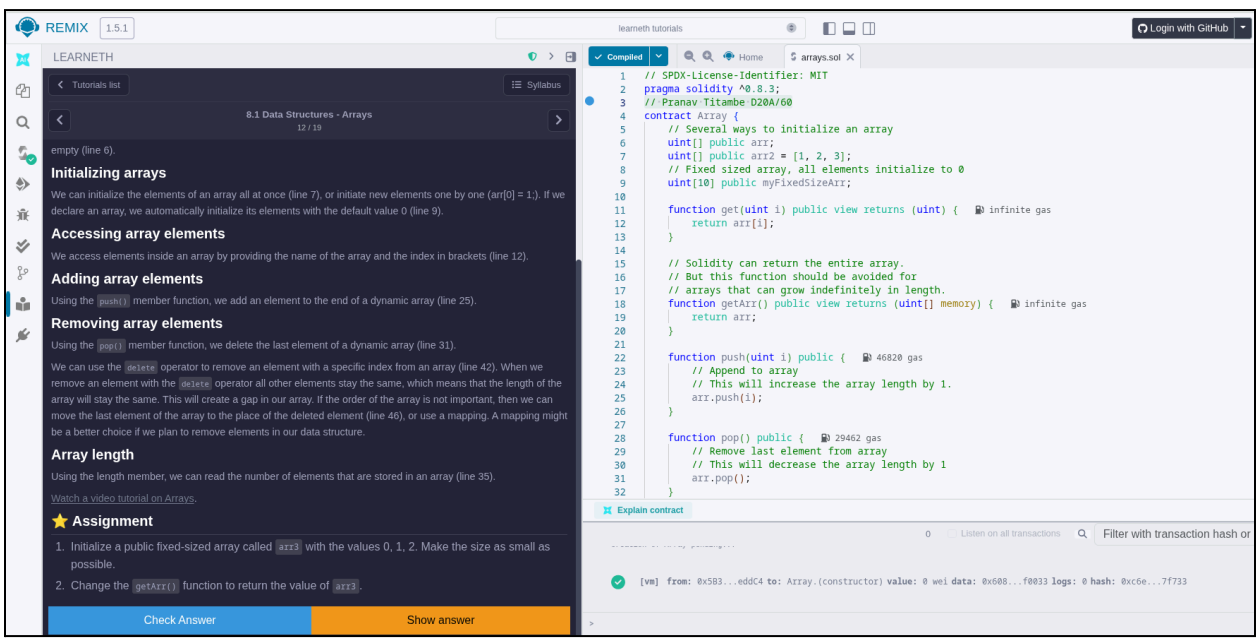

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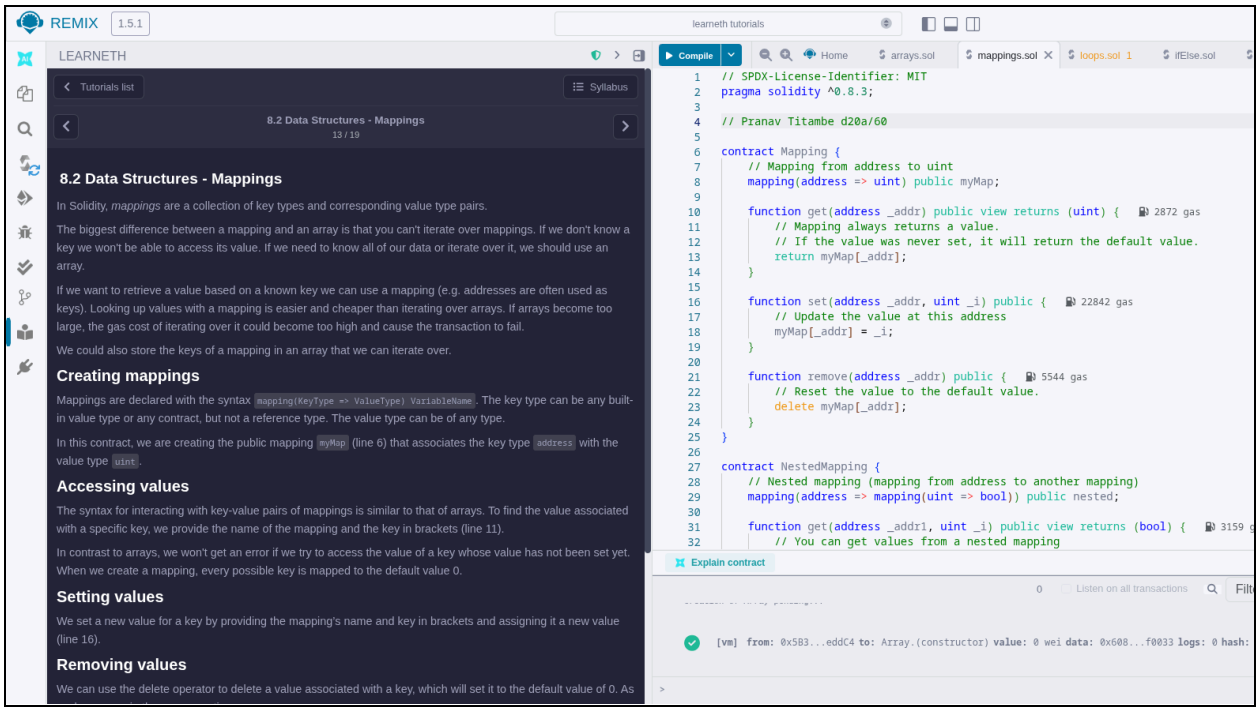

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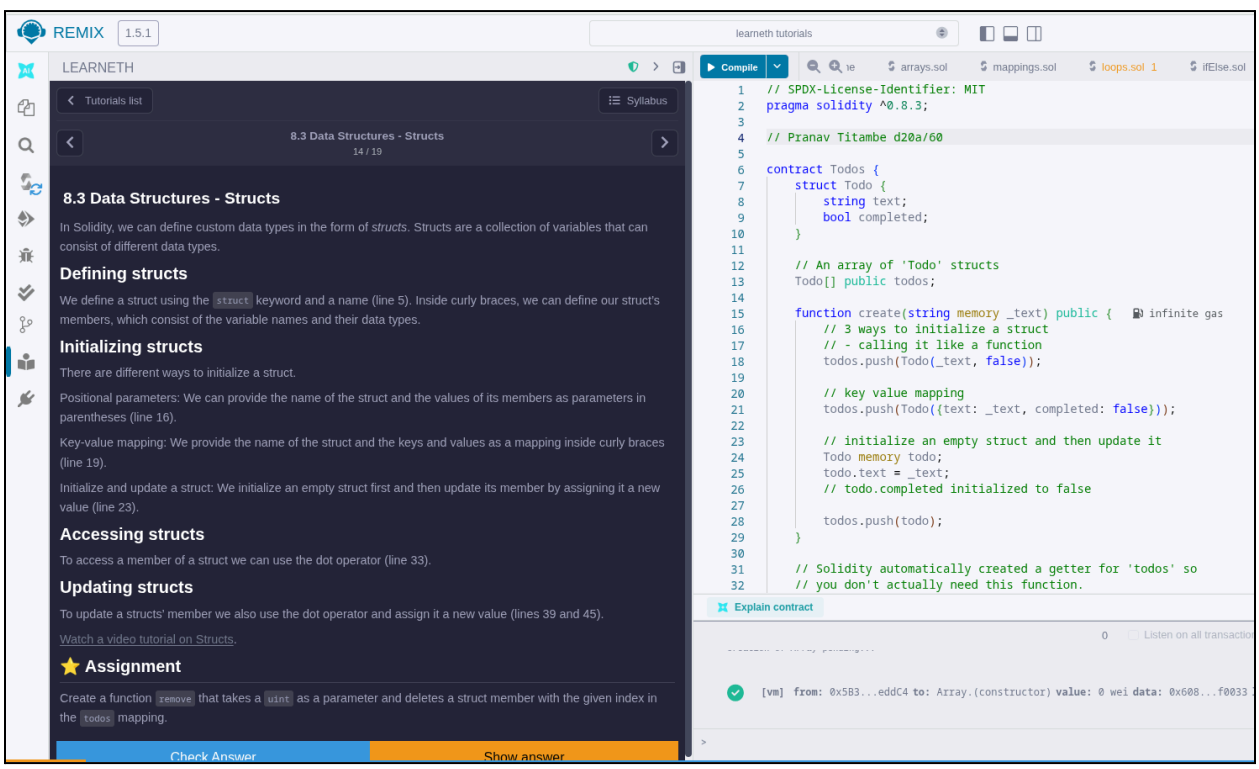

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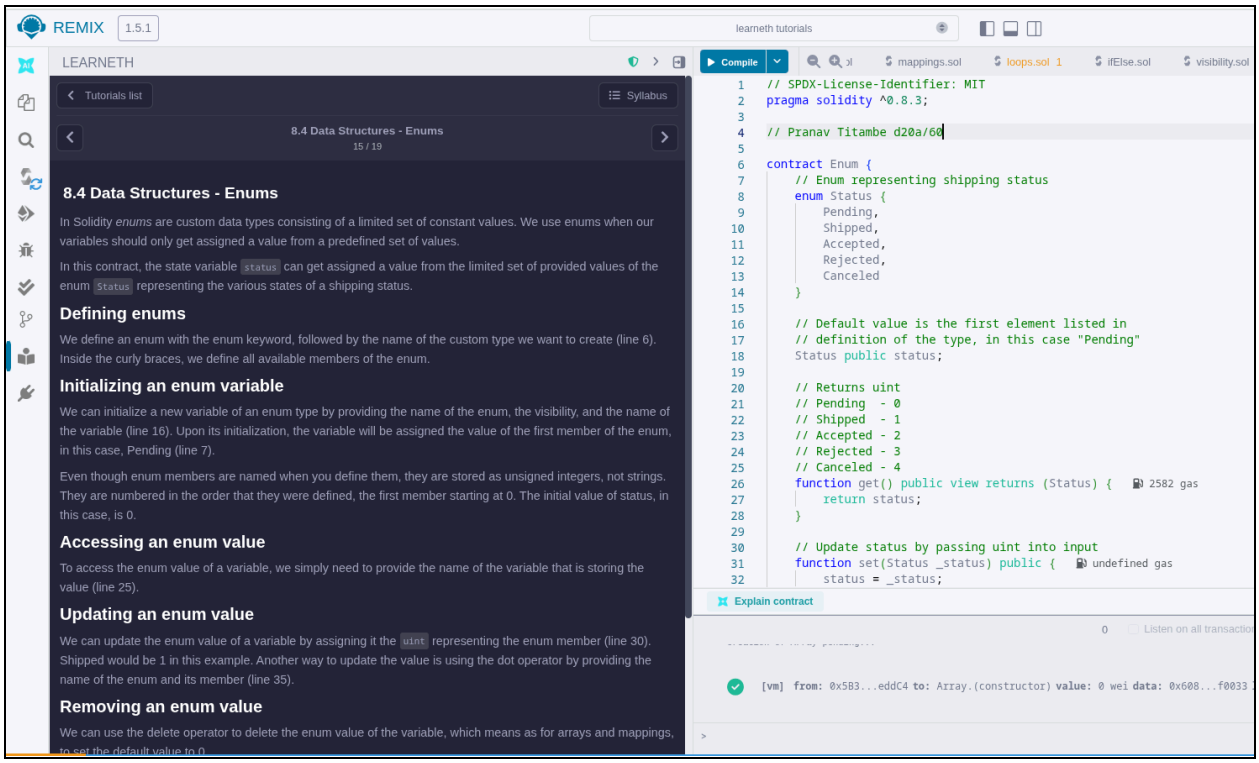

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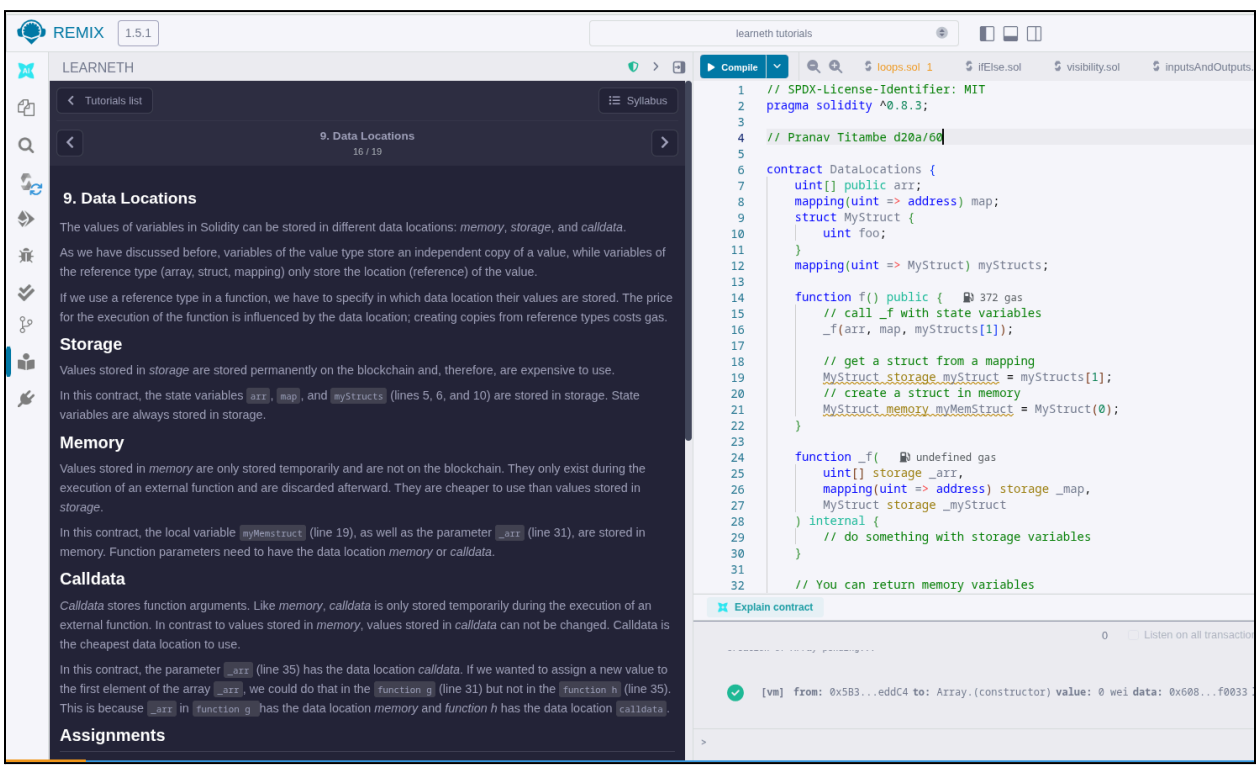

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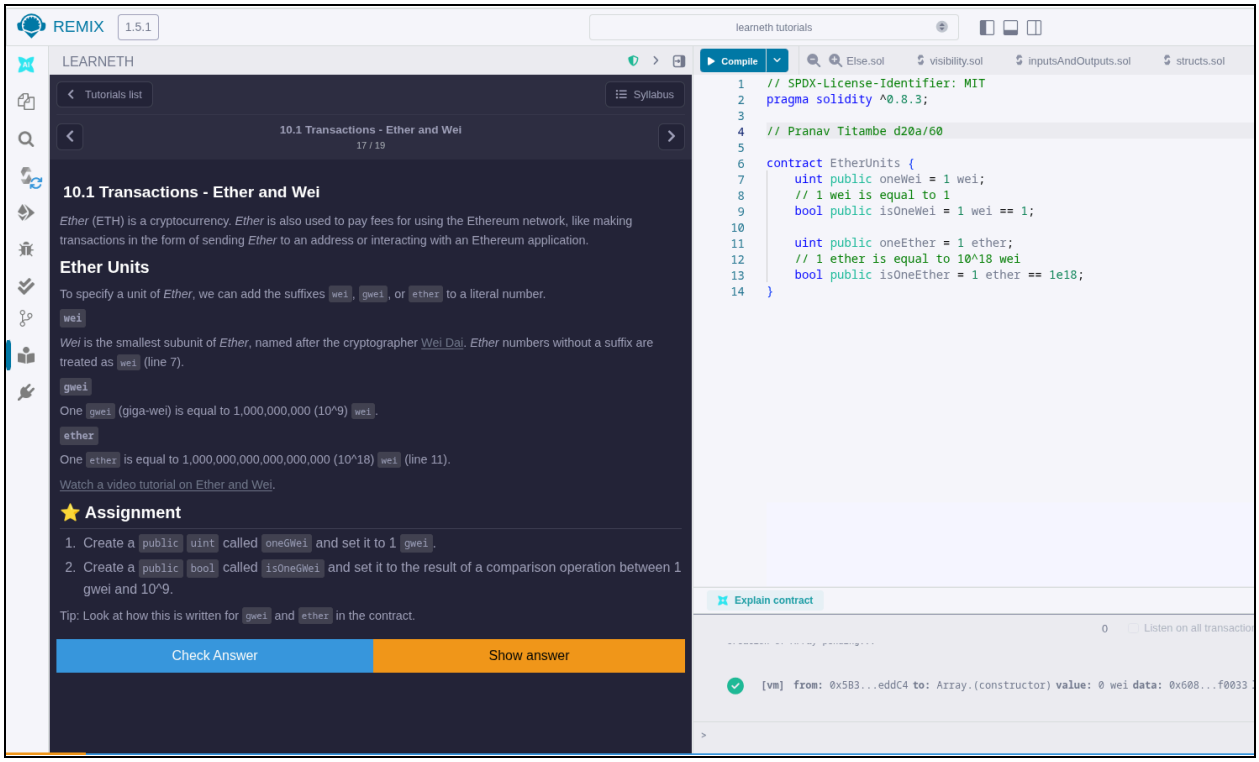

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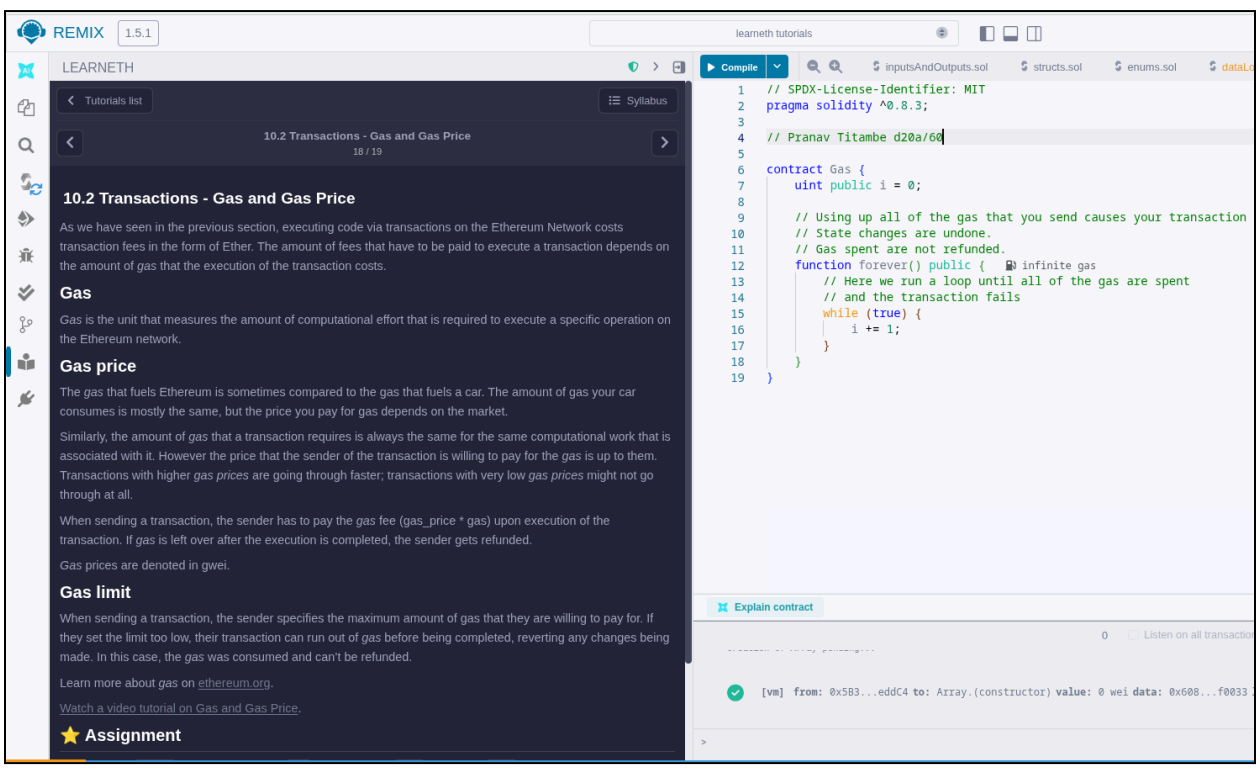

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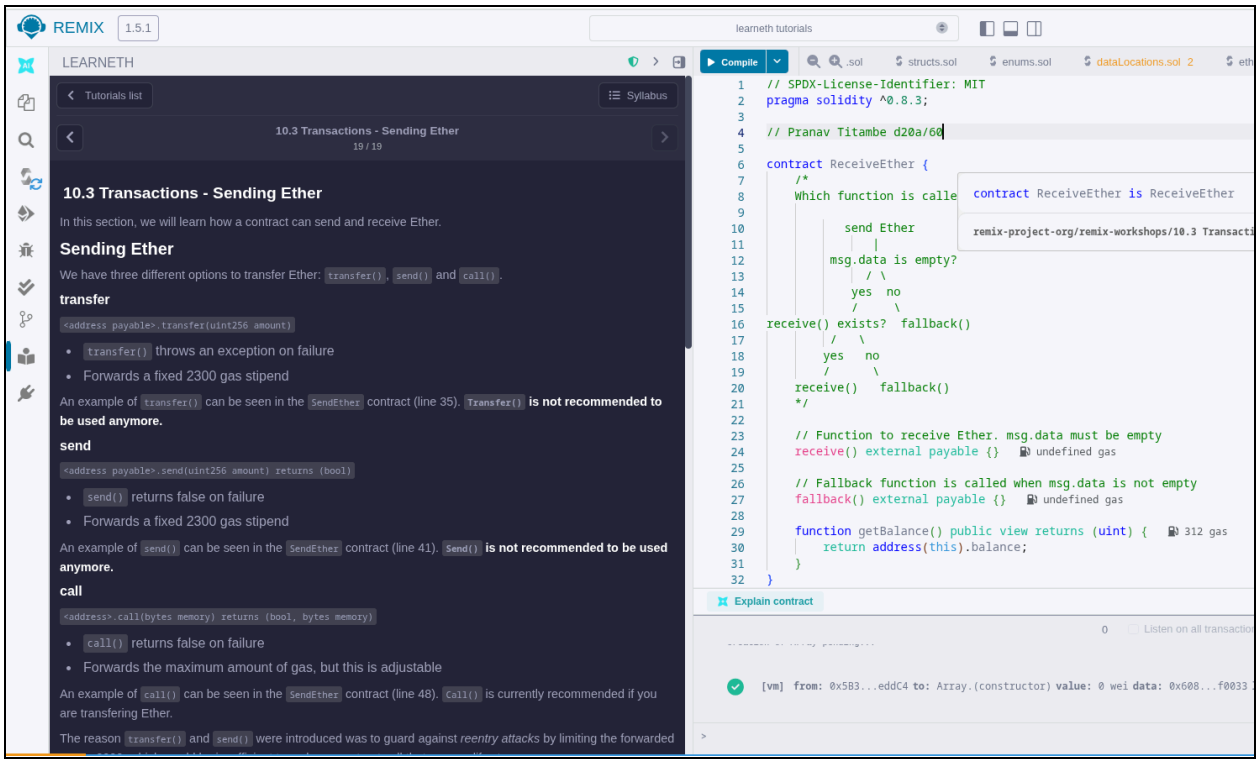

Conclusion: Through this experiment, the fundamentals of Solidity programming were explored by completing practical assignments in the Remix IDE. Concepts such as data types, variables, functions, visibility, modifiers, constructors, control flow, data structures, and transactions were implemented and understood. The hands-on practice helped in designing, compiling, and deploying smart contracts on the Remix VM, thereby strengthening the understanding of blockchain concepts. This experiment provided a strong foundation for developing and managing smart contracts efficiently.